# How Chatbots Work KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

**Word bank:** next word tokens true

**1. Finish the sentence.** One word each.

Peanut butter and jelly Once upon a time I am going to the Answers vary.

**2. Count what comes next.** The training text:

> I like rice. I like soccer. I like rice and beans. My friend likes soccer. My friend likes rice.

Finish the table. Count how many times each word comes next. ("end" means the end of a sentence.)

| After this word | These words came next (how many times) | Dice |
|---|---|---|
| I | like (3) | always "like" |
| like | rice (2), soccer (1) | 1 to 4: rice, 5 or 6: soccer |
| likes | soccer (1), rice (1) | 1 to 3: soccer, 4 to 6: rice |
| rice | end (2), and (1) | 1 to 4: end, 5 or 6: and |
| soccer | end (2) | always end |
| and | beans (1) | always "beans" |
| beans | end (1) | always end |
| My | friend (2) | always "friend" |
| friend | likes (2) | always "likes" |

The dice column gives more common words more numbers.

**3. Tiny chatbot, real chatbot.**

Both chatbots predict the next word. Neither one checks if a sentence is true.

A real chatbot reads tokens: words and pieces of words.

**4. The dice chatbot.** Start with **I** or **My**. Roll a die when there is a choice.

Sentence 1: Answers vary. Example: I like rice and beans.

Sentence 2: _____

Sentence 3: _____

Did you write a sentence that is **not** in the training text? Which one? My friend likes rice and beans.

**5. Talk about it.** Did the tiny chatbot understand "My friend likes rice and beans"?

The tiny chatbot **did** / **did not** understand, because Answers vary. It only counted.

_____

_____

**Go Further (Day 2).** The word-counting chatbot on screen.

With "always pick the most common," is the sentence the same every time? yes

A strange sentence it wrote: Answers vary.

**ELL** A sentence it wrote in my home language: Answers vary.